DATA SCIENCE MINOR PROJECT REPORT

PROJECT REPORT

Obesity Level Prediction

Classifying Obesity Levels from Eating Habits and Physical Condition using Machine Learning

Submitted by

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Section: K24P24CQ

Course Code: INT 324

Under the Guidance of

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DECLARATION

I, Al Adnan, student of Bachelor of Technology under the CSE/IT Discipline at Lovely Professional University, Punjab, hereby declare that this project report titled “Obesity Level Prediction: Classifying Obesity Levels from Eating Habits and Physical Condition using Machine Learning” documents the work contained in my submitted project files.

The report describes the dataset, preprocessing steps, models, evaluation results, visualizations, assumptions and limitations evidenced in the project source code (obesity_level_prediction.py) and the dataset file (ObesityDataSet.csv). The dataset is obtained from the UCI Machine Learning Repository, and the fact that a large share of its records is synthetically generated is stated openly and is not represented as direct real-world measurement.

All numerical results in this report were produced by executing the submitted code on the submitted dataset with a fixed random seed (random_state = 42).

Date: 31st October 2026

Registration No.: 12402657

Name of the Student: Al Adnan

CERTIFICATE

This is to certify that Al Adnan, bearing Registration No. 12402657, has completed the Data Science Minor Project titled “Obesity Level Prediction: Classifying Obesity Levels from Eating Habits and Physical Condition using Machine Learning” under my guidance and supervision.

To the best of my knowledge, the present report documents the student’s project implementation, analytical study and development effort.

Signature and Name of the Supervisor: Manpreet Kaur

Designation of the Supervisor: Assistant Professor

School of Computer Science and Engineering

Lovely Professional University, Phagwara, Punjab.

Date: 31st October 2026

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my project supervisor, Manpreet Kaur, and the faculty members of the Lovely School of Computer Science and Engineering for their guidance and academic support during this project. Their suggestions helped shape it into a structured exercise in data analysis and supervised machine learning.

I also acknowledge the UCI Machine Learning Repository for hosting the "Estimation of Obesity Levels Based on Eating Habits and Physical Condition" dataset, and Fabio Mendoza Palechor and Alexis De la Hoz Manotas, who created the dataset and described it in their 2019 paper in Data in Brief. Making such data openly available under a Creative Commons licence makes projects like this possible.

Finally, I acknowledge the open-source Python ecosystem, particularly Pandas, NumPy, scikit-learn and Matplotlib, which supports the data processing, modelling and visualization carried out in this project.

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1. INTRODUCTION

1.1 Background

Obesity is a long-term health condition in which excess body fat accumulates to a degree that can harm health. It is associated with higher risk of type 2 diabetes, cardiovascular disease, hypertension and several other conditions, and it is widely described as a growing public-health concern in many regions of the world. Body weight is influenced by many factors, including diet, physical activity, family history, age, sleep, and the environment a person lives in.

A common screening measure is the Body Mass Index (BMI), calculated as weight in kilograms divided by the square of height in metres. BMI ranges are used to place a person in categories such as underweight, normal weight, overweight and obesity classes I to III. BMI is simple to compute, but it requires measured height and weight. A natural data science question is therefore whether a person's obesity level can be estimated from habits and lifestyle information alone, such as eating patterns, water intake, physical activity and mode of transport.

This project studies that question using the UCI dataset "Estimation of Obesity Levels Based on Eating Habits and Physical Condition", which contains 2,111 records of individuals from Mexico, Peru and Colombia and a seven-class obesity-level label. Four supervised classification algorithms are trained and compared, and an additional experiment shows how the results change when height and weight are included as input features.

1.2 Problem Statement

The target variable in this dataset was derived from BMI. If height and weight are supplied to a model, it can simply re-learn the BMI formula, giving very high accuracy that says little about lifestyle. The real problem addressed here is: given only demographic information, eating habits and physical-condition variables, how well can a machine learning model classify a person into one of seven obesity levels, which algorithm performs best, and which behavioural features contribute most to the prediction?

1.3 Aim and Objectives

- Load, inspect and clean the UCI obesity-level dataset and document its structure, quality and class balance.
- Perform exploratory data analysis of numeric and categorical variables and their relationship with the obesity level.

- Prepare the data using label encoding, one-hot encoding, a stratified train/test split and feature scaling.
- Train and compare four classifiers: Logistic Regression, Decision Tree, Random Forest and K-Nearest Neighbors.
- Evaluate the models using accuracy, macro precision, macro recall, macro F1 score, per-class reports and confusion matrices.
- Identify the most influential features using Random Forest feature importance.
- Demonstrate the target-leakage effect by repeating the experiment with Height and Weight included.

1.4 Scope of the Project

The scope covers a single Python script (obesity_level_prediction.py) that performs data loading, cleaning, feature selection, encoding, model training, evaluation and visualization on the file ObesityDataSet.csv. A switch named USE_HEIGHT_WEIGHT controls whether Height and Weight are included. The default, False, represents the honest lifestyle-only task; setting it to True reproduces the leakage setting. The project does not include a deployed application, clinical validation or hyper-parameter search.

1.5 Significance

The project is educational and analytical. It demonstrates a complete supervised-learning workflow on a multi-class health-related dataset and illustrates two important practical lessons: the need to detect and avoid target leakage, and the need to read model accuracy in light of how the data were created. Because a majority of the dataset is synthetic, the results should be understood as a demonstration of method, not as a clinical tool.

1.6 Report Organization

Chapter 2 documents the dataset source and its attributes. Chapter 3 presents the exploratory data analysis and preprocessing. Chapter 4 describes the models, evaluation results, feature importance and the leakage experiment. Chapters 5 and 6 give the conclusion and future scope, and Chapter 7 lists the references. The appendices contain the file list, formulas, full classification reports and the execution environment.

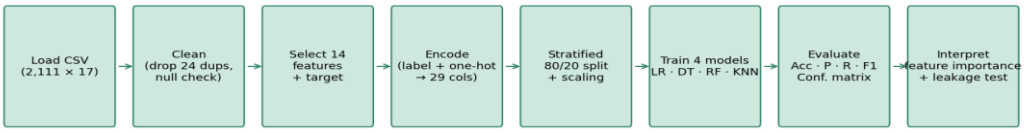


Figure 1. Analytical workflow of the project.

2. SOURCE OF DATASET

2.1 Dataset Source

The dataset is obtained from the UCI Machine Learning Repository, titled "Estimation of Obesity Levels Based On Eating Habits and Physical Condition" (dataset ID 544, donated on 26 August 2019, DOI 10.24432/C5H31Z). It was created by Fabio Mendoza Palechor and Alexis De la Hoz Manotas and described in the paper "Dataset for estimation of obesity levels based on eating habits and physical condition in individuals from Colombia, Peru and Mexico", published in Data in Brief in 2019. The repository lists it under the subject area Health and Medicine, for classification, regression and clustering tasks, and licenses it under Creative Commons Attribution 4.0 International (CC BY 4.0).

2.2 How the Data Were Created

According to the UCI documentation, 77% of the records were generated synthetically using the Weka tool and the SMOTE (Synthetic Minority Over-sampling Technique) filter, and 23% were collected directly from users through a web platform. The synthetic generation was used to balance the classes. This has important consequences that are discussed in Chapters 3 and 4: values of some ordinal survey variables appear as non-integers, some classes show unusual internal uniformity, and models can look better on a random split than they would on genuinely new people.

2.3 Dataset Overview

Property	Value in the supplied file
File used	ObesityDataSet.csv
Raw shape	2,111 rows × 17 columns
Shape after duplicate removal	2,087 rows × 17 columns (24 duplicate rows removed)
Missing values	0 in every column
Target variable	NObeyesdad (7 classes)
Input features used	14 (16 when Height and Weight are included)
Features after one-hot encoding	29 (31 with Height and Weight)
Population	Individuals from Mexico, Peru and Colombia; age 14 to 61 years
Licence	CC BY 4.0

2.4 Attribute Dictionary

The column codes in the file are abbreviations of the original Spanish-language survey items. Their meaning, as documented in the project code and the dataset paper, is given below.

Column	Type	Meaning
Gender	Categorical	Female or Male
Age	Continuous	Age in years (14 to 61)
Height	Continuous	Height in metres
Weight	Continuous	Weight in kilograms
family_history_with_overweight	Binary	Whether a family member has suffered from overweight
FAVC	Binary	Frequent consumption of high-calorie food
FCVC	Ordinal (1–3)	Frequency of vegetable consumption in meals
NCP	Ordinal (1–4)	Number of main meals per day
CAEC	Categorical	Eating between meals: no, Sometimes, Frequently, Always
SMOKE	Binary	Whether the person smokes
CH2O	Ordinal (1–3)	Daily water consumption
SCC	Binary	Whether the person monitors calorie intake
FAF	Ordinal (0–3)	Physical activity frequency
TUE	Ordinal (0–2)	Time spent using technology devices
CALC	Categorical	Alcohol consumption: no, Sometimes, Frequently, Always
MTRANS	Categorical	Usual transport: Public Transportation, Automobile, Walking, Motorbike, Bike
NObeyesdad	Target	Obesity level (seven classes)

2.5 The Target Variable

The target NObeyesdad has seven ordered classes: Insufficient_Weight, Normal_Weight, Overweight_Level_I, Overweight_Level_II, Obesity_Type_I, Obesity_Type_II and Obesity_Type_III. The dataset documentation states that the labels follow BMI thresholds (below 18.5 underweight, 18.5 to 24.9 normal, 25.0 to 29.9 overweight, 30.0 to 34.9 obesity I, 35.0 to 39.9 obesity II, and above 40 obesity III). Because the label is a deterministic function of Height and Weight, using those two columns as inputs causes target leakage. This is why the code keeps them out of the feature set by default.

2.6 Data Provenance Statement

The supplied CSV file is named ObesityDataSet.csv, while the file listed on the UCI page is named ObesityDataSet_raw_and_data_sinthetic.csv; the commented-out path in the source code refers to the latter name. The file loaded by the script has the same 2,111 rows and 17 columns described by UCI. Both real survey responses and SMOTE-generated records appear in the file without a flag column that distinguishes them, so the analysis cannot separate real from synthetic records.

3. EDA PROCESS

3.1 Purpose of Exploratory Data Analysis

Exploratory Data Analysis (EDA) is used to understand the structure, completeness, distribution and relationships in a dataset before applying a model. In this project EDA supports data validation, duplicate and missing-value checks, type inspection, class-balance assessment and visualization, and it helps identify which characteristics of the data need to be stated as caveats.

3.2 Data Loading and Structure

The script reads the file with pandas.read_csv and prints the shape, the first five rows and the column names. The file has 2,111 rows and 17 columns: eight numeric columns (Age, Height, Weight, FCVC, NCP, CH2O, FAF, TUE) and nine text columns (Gender, family_history_with_overweight, FAVC, CAEC, SMOKE, SCC, CALC, MTRANS and the target NObeyesdad).

3.3 Data Quality Checks

The cleaning step removes duplicate rows with drop_duplicates() and drops any row with a missing target. The run found 24 duplicate rows, which reduced the dataset from 2,111 to 2,087 records. By class, the 24 removed rows were 14 from Overweight_Level_I, 5 from Normal_Weight and 5 from Insufficient_Weight, so the class counts changed slightly (for example, Overweight_Level_I fell from 290 to 276). No column has missing values, so the imputation code in the script (a median fill for numeric fields and the label “Unknown” for categorical ones) is a safeguard that changes nothing on this file.

3.4 Target Distribution

After cleaning, the seven classes are well balanced, as expected for a dataset that was oversampled to balance classes. The largest class is Obesity_Type_I with 351 records (16.82%), and the smallest is Insufficient_Weight with 267 records (12.79%). The ratio between the largest and smallest class is about 1.3 to 1, so accuracy is a meaningful metric here, and macro-averaged metrics are used in addition to give each class equal weight.

Obesity level	Records	Share (%)	BMI range in data (min – max)
Insufficient_Weight	267	12.79	13.00 – 19.08
Normal_Weight	282	13.51	18.49 – 24.91
Overweight_Level_I	276	13.22	22.83 – 28.77
Overweight_Level_II	290	13.90	25.71 – 30.36

Obesity level	Records	Share (%)	BMI range in data (min – max)
Obesity_Type_I	351	16.82	29.91 – 35.17
Obesity_Type_II	297	14.23	34.05 – 39.79
Obesity_Type_III	324	15.52	36.77 – 50.81
Total	2,087	100.00	

BMI = Weight / Height², minimum and maximum within each class after duplicate removal.

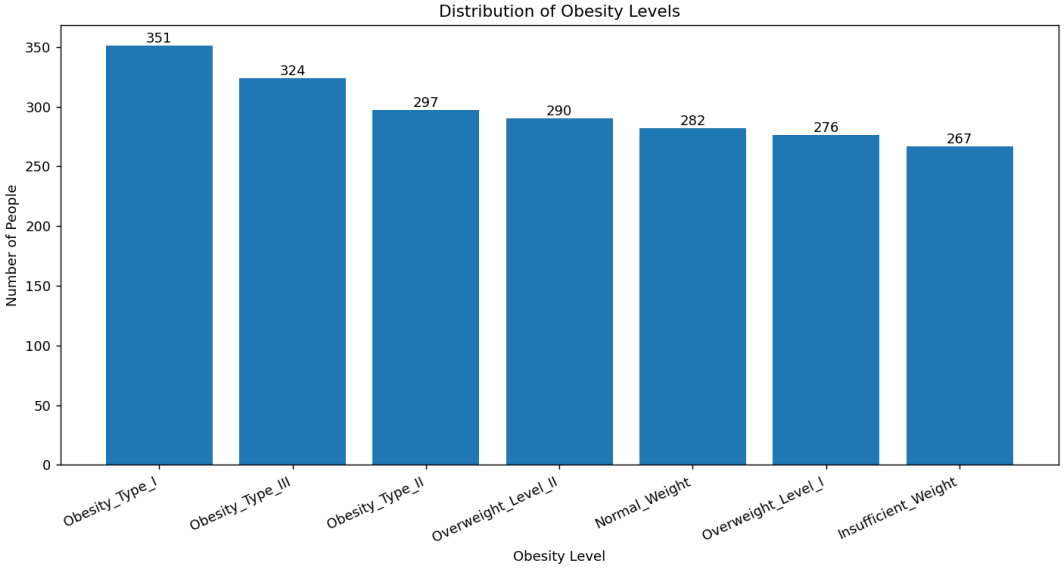


Figure 2. Distribution of obesity levels after duplicate removal (generated by the project script).

3.5 Numeric Variables

The numeric summary on the raw file shows a young sample: the mean age is 24.3 years and the median 22.8, with a maximum of 61. Height ranges from 1.45 m to 1.98 m and weight from 39 kg to 173 kg. The ordinal survey variables FCVC, CH2O, FAF and TUE and the count NCP are meant to take whole-number answers, but 39% to 61% of their values are non-integers (for example, 61% for CH2O and 57% for FAF). This is a visible trace of SMOTE interpolation between real respondents and is a reminder that this file is not a raw survey export.

Variable	Mean	Std. dev.	Min	Max	Non-integer share
Age	24.31	6.35	14	61	65%
Height (m)	1.70	0.09	1.45	1.98	–
Weight (kg)	86.59	26.19	39	173	–
FCVC	2.42	0.53	1	3	39%
NCP	2.69	0.78	1	4	30%
CH2O	2.01	0.61	1	3	61%
FAF	1.01	0.85	0	3	57%
TUE	0.66	0.61	0	2	55%

Statistics computed on the raw 2,111-row file before duplicate removal.

3.6 Age

Age is right-skewed, with most respondents between about 18 and 26 years. Obesity_Type_II has the highest mean age (28.2 years) and Insufficient_Weight the lowest (19.8 years). Age therefore carries a real signal, but it also reflects how the sample was built: the younger classes and the older classes come from somewhat different regions of the age range.

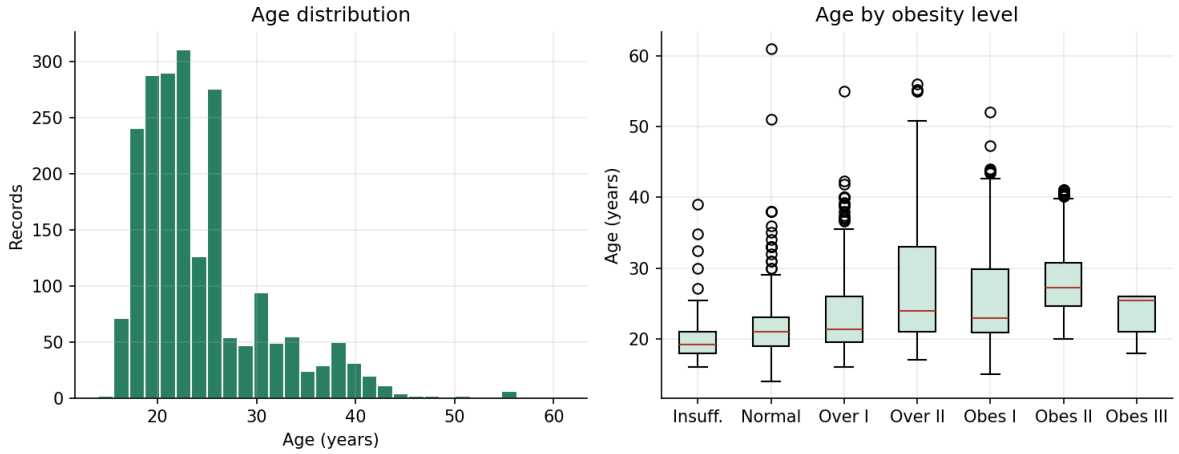


Figure 3. Age distribution (left) and age by obesity level (right).

3.7 Categorical Variables

Most respondents report a family history of overweight (1,726 of 2,111) and frequent high-calorie food (1,866). Very few smoke (44) and few monitor their calories (96). The great majority use public transportation (1,580) or a car (457), and only 56 walk. Snacking between meals is “Sometimes” for 1,765 respondents. The rare categories (for example Always for CALC, which occurs once, and Bike and Motorbike for MTRANS) offer almost no training signal and also produce sparse one-hot columns.

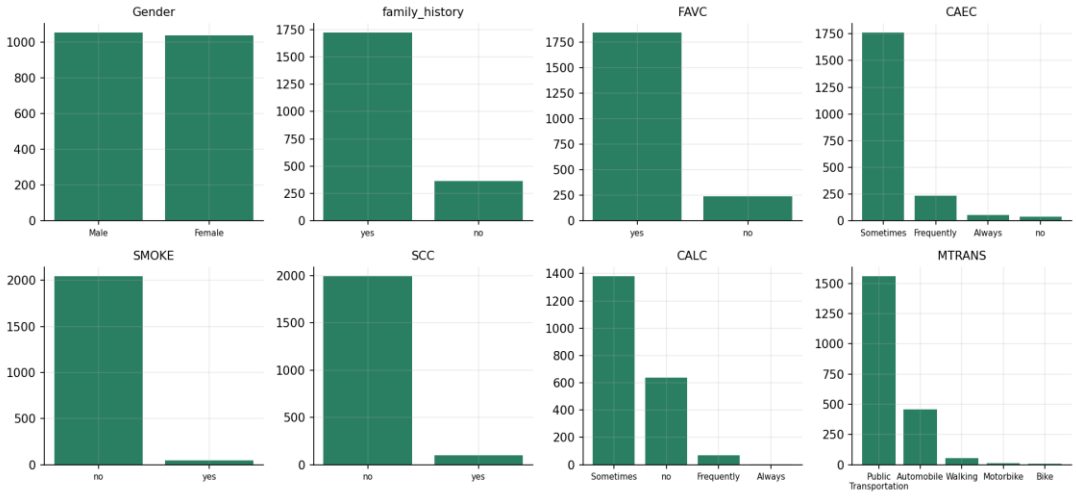


Figure 4. Distribution of the eight categorical variables.

3.8 Correlation Between Numeric Variables

Linear correlations among the numeric variables are weak to moderate. The strongest are Height with Weight (0.46), TUE with Age (−0.30), and Height with FAF (0.29). Among the habit variables, no pair exceeds 0.17 in absolute value. Low correlation between predictors is helpful for the linear model, but it also means no single numeric habit separates the classes by itself.

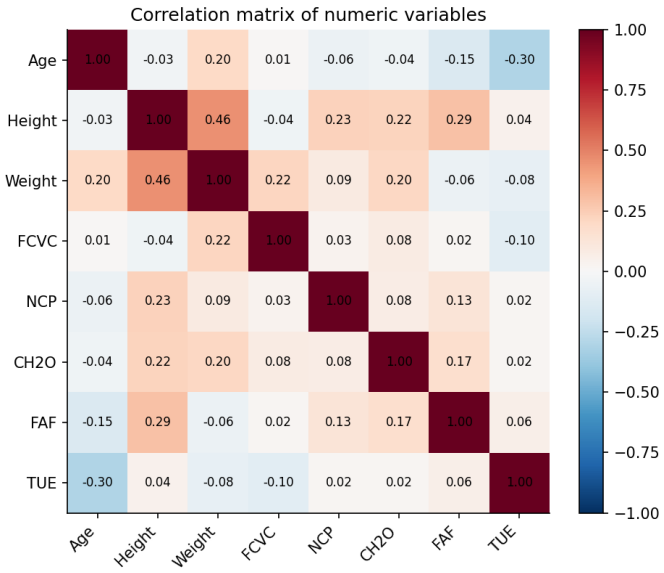


Figure 5. Correlation matrix of numeric variables.

3.9 Relationship Between BMI and the Label

Computing BMI as Weight / Height² and plotting it by class confirms that the label is derived from BMI: the classes occupy successive, mostly non-overlapping BMI bands. A few records sit outside the nominal cut-offs (for example, the minimum BMI in Obesity_Type_III is 36.8 and the maximum in Overweight_Level_II is 30.4), so the label is close to, but not exactly, a strict threshold rule. This analysis is the justification for the leakage experiment in Section 4.9.

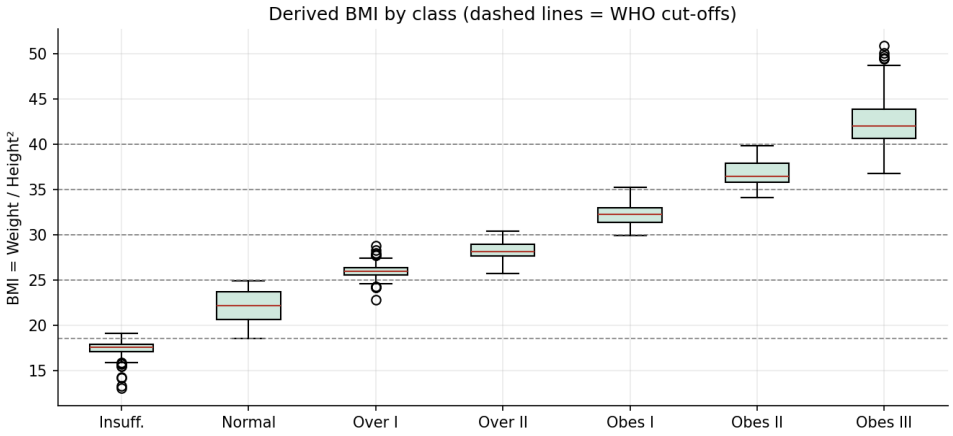


Figure 6. BMI by obesity level. Dashed lines mark the BMI cut-offs 18.5, 25, 30, 35 and 40.

3.10 Gender and the Synthetic Data Artefact

Gender composition is balanced overall (1,068 male and 1,043 female), but it is extremely unbalanced within two classes: Obesity_Type_III has 323 females and one male, while Obesity_Type_II has 295 males and two females. Obesity_Type_III is also unusual in that every record has FCVC = 3 and NCP = 3, a mean age of 23.5 years and a maximum age of 26. These patterns are unlikely to describe real populations. They are most plausibly a result of how the synthetic records were generated from a small number of real respondents, and they explain why Obesity_Type_III is the easiest class for every model in Chapter 4.

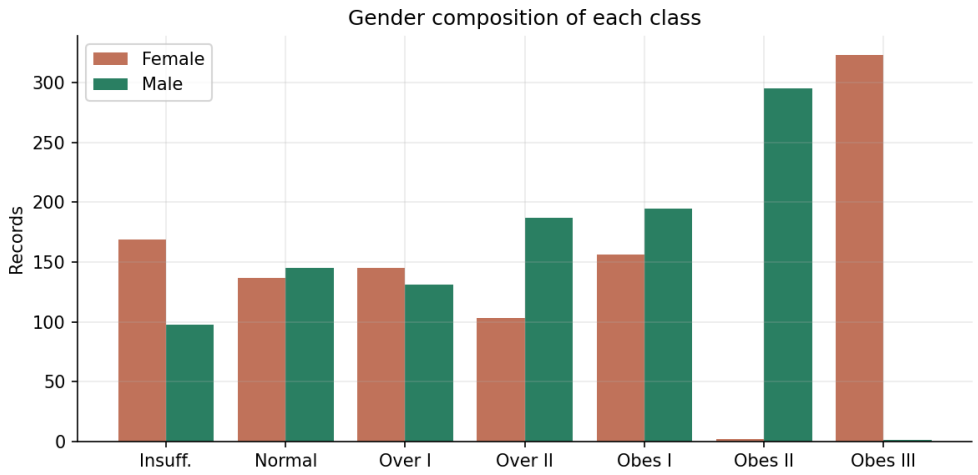


Figure 7. Gender composition of each obesity level.

3.11 Preprocessing Steps Implemented

- Feature selection: 14 columns are used as inputs (Gender, Age, family_history_with_overweight, FAVC, FCVC, NCP, CAEC, SMOKE, CH2O, SCC, FAF, TUE, CALC, MTRANS); Height and Weight are excluded by default.
- Target encoding: LabelEncoder maps the seven class names to the integers 0 to 6 in alphabetical order (0 Insufficient_Weight, 1 Normal_Weight, 2 Obesity_Type_I, 3 Obesity_Type_II, 4 Obesity_Type_III, 5 Overweight_Level_I, 6 Overweight_Level_II).
- Feature encoding: pandas.get_dummies one-hot encodes the eight categorical columns with drop_first=False, expanding 14 columns into 29.
- Train/test split: 80% training (1,669 records) and 20% testing (418 records), stratified by class, with random_state = 42.
- Scaling: StandardScaler is fitted on the training set only and applied to the test set; the scaled data are used for Logistic Regression and KNN, while tree-based models use the unscaled data.

3.12 EDA Findings and Data Caveats

- The dataset has no missing values and 24 exact duplicate rows, which were removed.
- The seven classes are reasonably balanced (12.8% to 16.8% each).
- About 77% of the records are synthetic, so test results on a random split can be optimistic because synthetic points can be near neighbours of the real points they were generated from.
- Several ordinal survey variables hold fractional values, and two classes are almost single-gender, which are signs of synthetic generation.
- The label is derived from BMI, so Height and Weight must be excluded for a meaningful lifestyle-based prediction.
- The data cover young adults from three Latin American countries and cannot be generalised to other populations or age groups.

3.13 EDA Workflow Summary

The implemented workflow is: load data → inspect schema → drop duplicates and check missing values → inspect the target distribution → select features → encode target and categorical variables → split the data with stratification → scale for distance-based and linear models → train four classifiers → evaluate → interpret feature importance → repeat with Height and Weight to expose leakage.

4. ANALYSIS ON DATASET

4.1 Experimental Design and the Leakage Setting

The task is a seven-class classification problem. All four models are trained on the same 1,669 training records and evaluated on the same 418 test records. The main experiment (USE_HEIGHT_WEIGHT = False) uses lifestyle and demographic variables only. A second run (USE_HEIGHT_WEIGHT = True) adds Height and Weight to show the leakage effect.

Because the classes are balanced, accuracy is reported together with macro-averaged precision, recall and F1. Macro averaging computes each metric per class and then averages them with equal weights. Models are ranked by macro F1 score. Three of the models use class_weight="balanced", which adjusts for any remaining imbalance; KNN has no class-weight option.

Model	Input data	Settings
Logistic Regression	Scaled	max_iter = 2000, class_weight = "balanced"
Decision Tree	Unscaled	max_depth = 10, min_samples_split = 10, class_weight = "balanced", random_state = 42
Random Forest	Unscaled	n_estimators = 200, max_depth = 15, min_samples_split = 5, class_weight = "balanced", random_state = 42
K-Nearest Neighbors	Scaled	n_neighbors = 5

4.2 Logistic Regression

Logistic Regression is a linear baseline that estimates class probabilities as a softmax of a linear combination of the features. It is fast and interpretable but can only form linear class boundaries. On the lifestyle-only task it reached 0.6316 accuracy and 0.5981 macro F1, the weakest of the four models. It separated Obesity_Type_III very well (F1 0.96) but handled Overweight_Level_II very poorly, with a recall of 0.12: only 7 of 58 test cases were found, and most were assigned to Obesity_Type_I or Obesity_Type_II. A linear model cannot easily represent the interaction between age, sex and habits that the neighbouring classes seem to need.

4.3 Decision Tree

A decision tree splits the feature space recursively with simple threshold rules. The tree here is limited to depth 10 with a minimum of 10 samples to split, which controls overfitting. It reached 0.6914 accuracy and 0.6879 macro F1. Its per-class results are more even than Logistic

Regression: the lowest F1 is 0.54 (Normal_Weight), and it classifies Obesity_Type_III almost perfectly (F1 0.98).

Figure 8 shows decision boundaries of a depth-5 tree trained on only two features, Age and FAF, to illustrate how the feature space is partitioned. Overlapping colours show that two features alone cannot separate seven classes. Figure 9 shows the top three levels of the depth-10 tree used for evaluation.

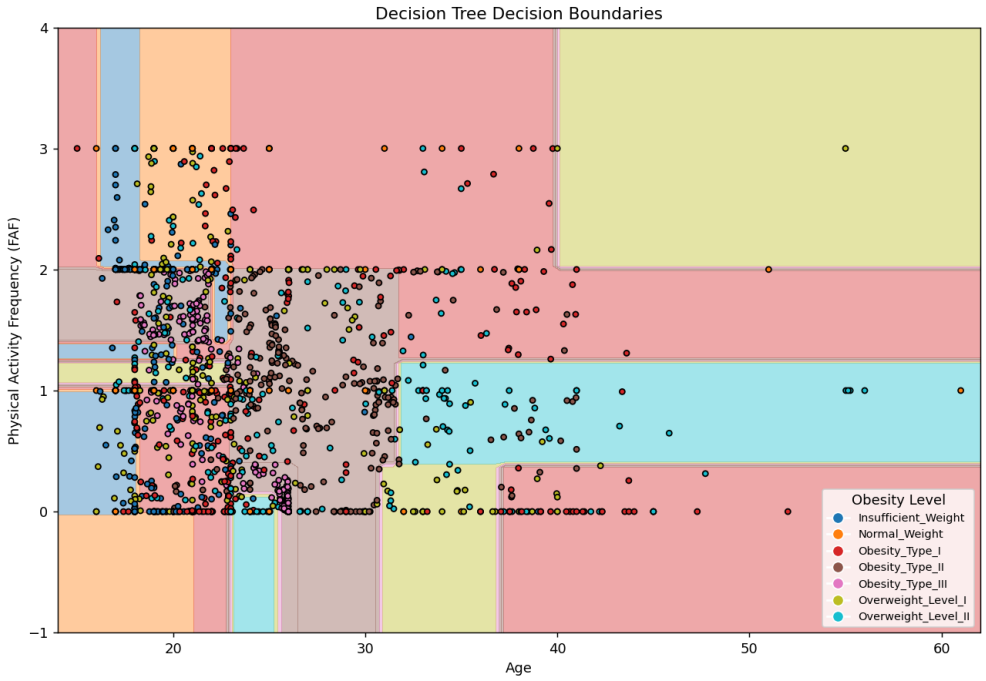


Figure 8. Decision tree decision boundaries on Age and FAF (illustration).

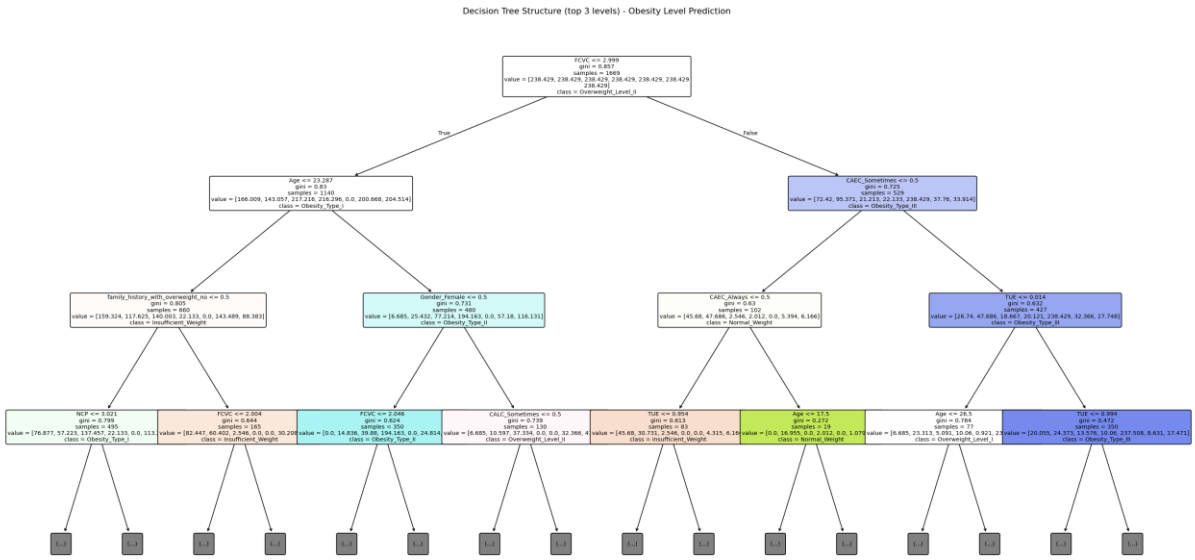


Figure 9. Decision tree structure (top three levels).

4.4 Random Forest

Random Forest averages many decorrelated decision trees, each trained on a bootstrap sample with random feature subsets. This reduces the variance of a single tree and can capture non-linear interactions between features. With 200 trees and a maximum depth of 15 it achieved the best result on the lifestyle-only task: 0.8397 accuracy, 0.8400 macro precision, 0.8366 macro recall and 0.8368 macro F1. Every class has an F1 of at least 0.71, and Obesity_Type_III reaches 0.99. The weakest classes are Normal_Weight (F1 0.71) and Overweight_Level_II (0.73), which lie next to each other on the BMI scale.

4.5 K-Nearest Neighbors

KNN classifies a case by the majority label among its five nearest training cases in the scaled feature space. It reached 0.7608 accuracy and 0.7500 macro F1, second among the four models. Because KNN depends on distance, scaling was applied. KNN is also the model most likely to benefit from SMOTE-generated data, since synthetic points lie between real points, so some of its performance may reflect that structure rather than real predictive power.

4.6 Model Comparison

Table below ranks the models by macro F1 on the 418 test records of the main experiment. The Random Forest leads on every metric, and the gap to the second-best model (KNN) is about 8 percentage points of accuracy and 8.7 points of macro F1. The ordering Random Forest > KNN > Decision Tree > Logistic Regression suggests that the relationship between habits and obesity level is strongly non-linear and interaction-based.

Model	Accuracy	Macro precision	Macro recall	Macro F1
Random Forest	0.8397	0.8400	0.8366	0.8368
KNN	0.7608	0.7542	0.7552	0.7500
Decision Tree	0.6914	0.6918	0.6857	0.6879
Logistic Regression	0.6316	0.6089	0.6233	0.5981

Table: model performance on the test set, USE_HEIGHT_WEIGHT = False (14 input features, 29 after encoding).

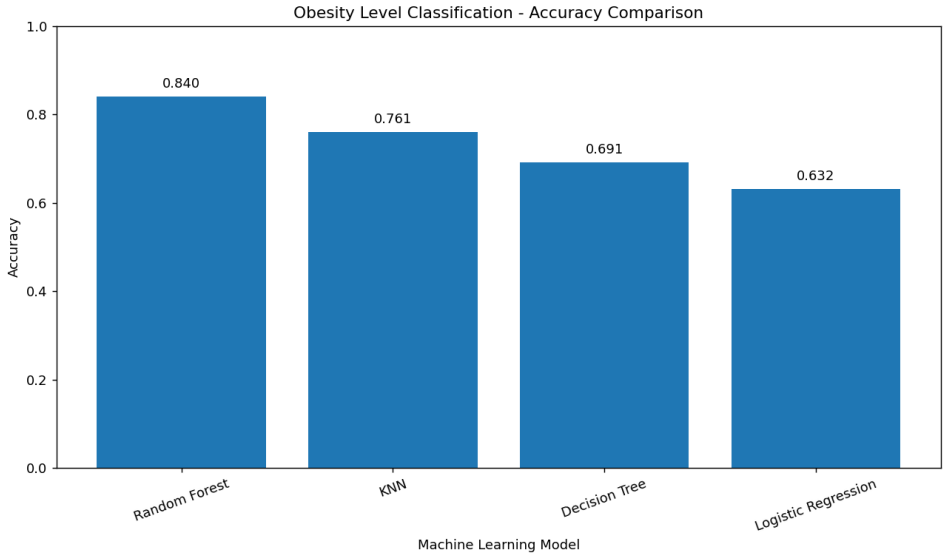


Figure 10. Accuracy comparison of the four models.

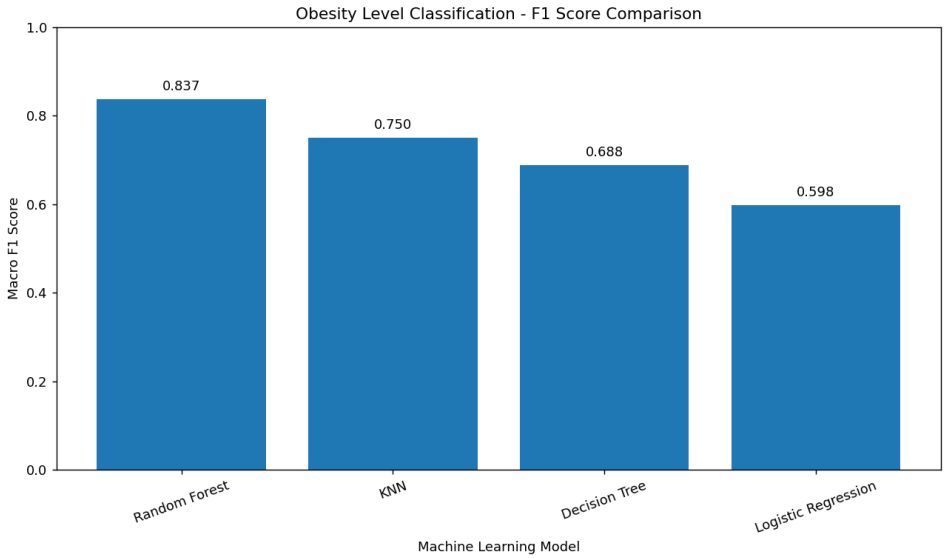


Figure 11. Macro F1 comparison of the four models.

One caution applies to these numbers: they come from a single random split with a fixed seed. With 418 test cases, one more or fewer correct prediction changes accuracy by about 0.24 percentage points, and a different seed could move the results slightly. Cross-validation would give a more stable estimate.

4.7 Confusion Matrix Analysis

The confusion matrices in Figure 12 show where each model makes mistakes. For all four models, Obesity_Type_III is almost always right (Random Forest: 64 of 65). Errors are concentrated among neighbouring classes: Normal_Weight is confused with Insufficient_Weight and with the two overweight levels, and the overweight levels are confused with Obesity_Type_I and with each other. This is an expected pattern, because an

ordinal scale with adjacent classes has fuzzy boundaries, and lifestyle alone cannot pin down a BMI band exactly. Random Forest has the strongest diagonal; its off-diagonal counts are small (at most 6, for Overweight_Level_I predicted as Normal_Weight and for Overweight_Level_II predicted as Normal_Weight or Obesity_Type_I). Logistic Regression has the clearest failure: 18 Overweight_Level_II cases predicted as Obesity_Type_I and 19 as Obesity_Type_II.

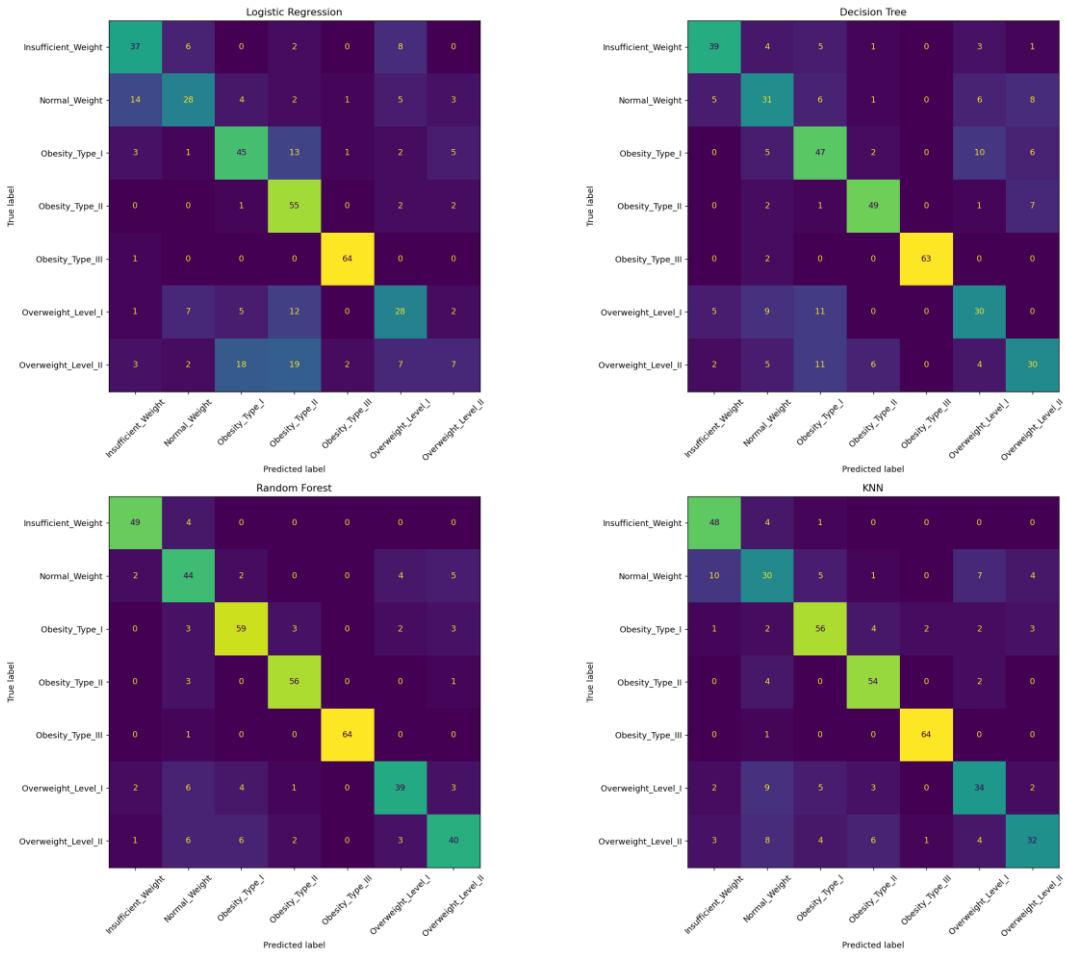


Figure 12. Confusion matrices of the four models on the test set.

4.8 Feature Importance

Random Forest feature importance measures how much each feature reduces impurity across the trees. In the lifestyle-only model, Age is the most important feature (0.146), followed by FCVC, vegetable consumption (0.125), NCP, the number of main meals (0.092), FAF, physical activity (0.081), TUE, technology use (0.080) and CH2O, water intake (0.078). Together these six numeric variables account for about 60% of total importance. Gender, alcohol consumption, family history, snacking and transport mode follow with smaller values, and calorie monitoring (SCC) contributes almost nothing (0.008 per dummy column).

These importances describe what the model used, not what causes obesity. Age and FCVC are important partly because of how the data were generated: Obesity_Type_III has FCVC = 3 for

every record and a narrow age range, and Obesity_Type_II has a much older and nearly all-male profile. Impurity-based importance is also known to favour continuous and high-cardinality variables over binary ones. The ranking is therefore a guide for further investigation and should not be read as a statement about health risk.

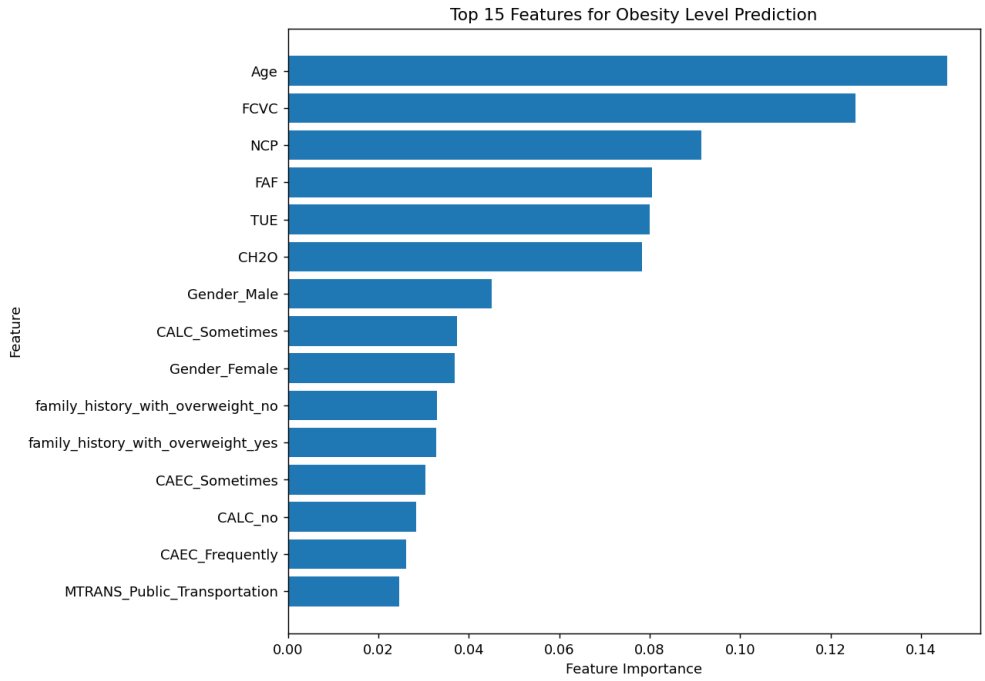


Figure 13. Top 15 Random Forest feature importances (lifestyle-only model).

4.9 Height and Weight Leakage Experiment

To demonstrate leakage, the entire pipeline was re-run with USE_HEIGHT_WEIGHT = True. Accuracy rose sharply for three of the four models, and the ranking changed completely. The Decision Tree reached 0.9354 accuracy (macro F1 0.9340), Random Forest 0.9306 (0.9292) and Logistic Regression 0.8923 (0.8882). KNN rose less, to 0.8086 (0.7989), possibly because its distance calculation is affected by the many one-hot columns. In the Random Forest, Weight became the single most important feature (0.291) and Height ranked fourth (0.076), displacing the lifestyle variables.

Model	Habits only	With Height & Weight	Change in accuracy
Logistic Regression	0.6316	0.8923	+0.2607
Decision Tree	0.6914	0.9354	+0.2440
Random Forest	0.8397	0.9306	+0.0909
KNN	0.7608	0.8086	+0.0478

Table: test accuracy with and without Height and Weight.

The added accuracy comes from the model re-learning the BMI rule that created the labels, not from understanding behaviour. The accuracy is still below 100% because trees approximate a

ratio (Weight / Height²) with axis-aligned splits and because the labels do not follow the BMI thresholds perfectly. For this reason the lifestyle-only figures in Section 4.6 are the honest estimate of predictive ability from habits, and the 0.84 accuracy of the Random Forest should be regarded as the headline result of this project.

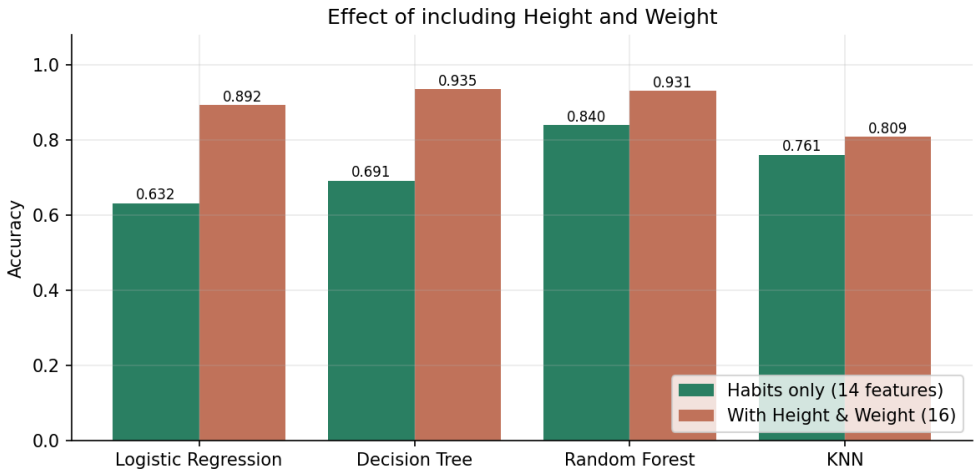


Figure 14. Effect of including Height and Weight on test accuracy.

4.10 Implementation and Reproducibility

The whole analysis is implemented in one Python script of about 25 sections with printed outputs at every stage. It uses pandas and NumPy for data handling, scikit-learn for preprocessing, models and metrics, and Matplotlib for plots. Random seeds are fixed (random_state = 42), so reruns on the same library versions give identical numbers. Recommendations for strengthening reproducibility:

- Record the Python and package versions (see Appendix D).
- Use k-fold cross-validation and report mean and standard deviation of the metrics.
- Tune hyper-parameters with a validation set or GridSearchCV instead of fixed values.
- Keep the leakage switch visible so both settings can be reproduced.
- Where possible, evaluate on real respondents only, held out before any oversampling.

5. CONCLUSION

This project built and evaluated a machine learning pipeline that classifies individuals into seven obesity levels from eating habits and physical-condition variables in the UCI obesity dataset. After removing 24 duplicates, 2,087 records were split into 1,669 training and 418 test records, and four classifiers were compared.

On the honest lifestyle-only task, the Random Forest performed best with 0.8397 accuracy and 0.8368 macro F1, followed by KNN (0.7608), the Decision Tree (0.6914) and Logistic Regression (0.6316). The strongest predictors were Age, vegetable consumption, number of main meals, physical activity, technology use and water intake. Mistakes were concentrated between adjacent classes, particularly Normal_Weight and the overweight levels, while Obesity_Type_III was recognised almost perfectly by every model.

Including Height and Weight lifted accuracy to about 93% for tree-based models, but this is leakage: the label was derived from those two variables through BMI. A central lesson of the project is therefore methodological: a high score must be checked against how the target was built.

A second lesson concerns data provenance. About 77% of the dataset is synthetic, and visible artefacts (fractional answers to ordinal questions, one-gender classes, constant values within a class) suggest that the results may be more optimistic than they would be on new people. The project's outputs should be read as a demonstration of method and relative model comparison, not as a screening tool or evidence about the causes of obesity.

6. FUTURE SCOPE

Cross-validation and tuning

Use stratified k-fold cross-validation and a systematic hyper-parameter search to obtain more stable and fairer comparisons.

Boosting models

Add gradient boosting, XGBoost or LightGBM, and compare them with the Random Forest.

Real-data evaluation

Hold out real (non-synthetic) respondents before oversampling, or collect a new real sample, to measure the true generalisation performance.

Ordinal modelling

Treat the seven classes as ordered and use ordinal classification or regression on BMI, with error measures that penalise distant mistakes more than adjacent ones.

Class-specific analysis

Study the weakest classes (Normal_Weight and the overweight levels) with targeted features or binary sub-models.

Explainability

Use SHAP values or permutation importance, which are more reliable than impurity-based importance, to explain predictions.

Fairness and bias

Audit performance across gender and age groups, given the near single-gender composition of two classes.

Deployment

Build a small web form where a person enters habits and receives a predicted category with clear disclaimers that it is educational and not medical advice.

7. REFERENCES

1. UCI Machine Learning Repository. Estimation of Obesity Levels Based On Eating Habits and Physical Condition [Dataset]. (2019). https://doi.org/10.24432/C5H31Z (https://archive.ics.uci.edu/dataset/544)

2. pandas development team. pandas documentation. https://pandas.pydata.org/docs/

3. NumPy developers. NumPy documentation. https://numpy.org/doc/

4. scikit-learn developers. scikit-learn documentation. https://scikit-learn.org/stable/

5. Project source files supplied for this report: obesity_level_prediction.py and ObesityDataSet.csv.

APPENDIX A: FILES AND CODE STRUCTURE

The project consists of one script and one data file.

- obesity_level_prediction.py: complete analysis script.
- ObesityDataSet.csv: 2,111 × 17 dataset from the UCI Machine Learning Repository.

The script is organised in these numbered sections:

Step	Content
1–2	Import libraries; load ObesityDataSet.csv and print shape, head and columns
3	Data cleaning: remove duplicates and rows with missing target
4–5	Select features and target; print class distribution
6–8	Handle missing values; label-encode target; one-hot encode categorical features
9–10	Stratified 80/20 train-test split; StandardScaler on training data
11–14	Train and evaluate Logistic Regression, Decision Tree (with boundary and tree plots), Random Forest and KNN
15–18	Compare models by macro F1 and identify the best model
19–21	Accuracy chart, F1 chart and confusion matrices
22–24	Random Forest feature importance and class distribution plot
25	Final analysis summary

APPENDIX B: FORMULA SUMMARY

Metric / calculation	Formula / description
BMI	Weight (kg) / Height (m)²
Accuracy	(TP + TN) / (TP + TN + FP + FN); for multi-class, correct predictions / total predictions
Precision (per class)	TP / (TP + FP)
Recall (per class)	TP / (TP + FN)
F1 score (per class)	2 × Precision × Recall / (Precision + Recall)
Macro average	Unweighted mean of the per-class metric over the 7 classes
Standardization	z = (x − mean) / standard deviation, fitted on training data only
KNN prediction	Majority class among the k = 5 nearest training points (Euclidean distance)
Gini impurity	$1 - \Sigma\, p_i^2$, where p_i is the proportion of class i in a node

APPENDIX C: PER-CLASS CLASSIFICATION REPORTS

Per-class precision, recall and F1 on the 418 test records for the lifestyle-only experiment (support in the last column).

Logistic Regression

Class	Precision	Recall	F1	Support
Insufficient_Weight	0.63	0.70	0.66	53
Normal_Weight	0.64	0.49	0.55	57
Obesity_Type_I	0.62	0.64	0.63	70
Obesity_Type_II	0.53	0.92	0.67	60
Obesity_Type_III	0.94	0.98	0.96	65
Overweight_Level_I	0.54	0.51	0.52	55
Overweight_Level_II	0.37	0.12	0.18	58

Decision Tree

Class	Precision	Recall	F1	Support
Insufficient_Weight	0.76	0.74	0.75	53
Normal_Weight	0.53	0.54	0.54	57
Obesity_Type_I	0.58	0.67	0.62	70
Obesity_Type_II	0.83	0.82	0.82	60
Obesity_Type_III	1.00	0.97	0.98	65
Overweight_Level_I	0.56	0.55	0.55	55
Overweight_Level_II	0.58	0.52	0.55	58

Random Forest

Class	Precision	Recall	F1	Support
Insufficient_Weight	0.91	0.92	0.92	53
Normal_Weight	0.66	0.77	0.71	57
Obesity_Type_I	0.83	0.84	0.84	70
Obesity_Type_II	0.90	0.93	0.92	60
Obesity_Type_III	1.00	0.98	0.99	65
Overweight_Level_I	0.81	0.71	0.76	55
Overweight_Level_II	0.77	0.69	0.73	58

K-Nearest Neighbors

Class	Precision	Recall	F1	Support
Insufficient_Weight	0.75	0.91	0.82	53

Class	Precision	Recall	F1	Support
Normal_Weight	0.52	0.53	0.52	57
Obesity_Type_I	0.79	0.80	0.79	70
Obesity_Type_II	0.79	0.90	0.84	60
Obesity_Type_III	0.96	0.98	0.97	65
Overweight_Level_I	0.69	0.62	0.65	55
Overweight_Level_II	0.78	0.55	0.65	58

APPENDIX D: EXECUTION ENVIRONMENT

The reported results were produced by running obesity_level_prediction.py with the following environment.

Item	Value
Python	3.12.3
pandas	3.0.2
NumPy	2.4.4
scikit-learn	1.8.0
Matplotlib	3.10.8
Random seed	42 (split, Decision Tree, Random Forest)
Settings	USE_HEIGHT_WEIGHT = False (main results); True (leakage experiment)

Exact figures may differ slightly in other library versions because of changes in default behaviour of the algorithms.

APPENDIX E: GRAPHS FROM THE HEIGHT AND WEIGHT RUN

The figures below are generated by the same script with USE_HEIGHT_WEIGHT = True. They accompany the leakage experiment in Section 4.9. The class-distribution plot is identical to Figure 2 and is not repeated.

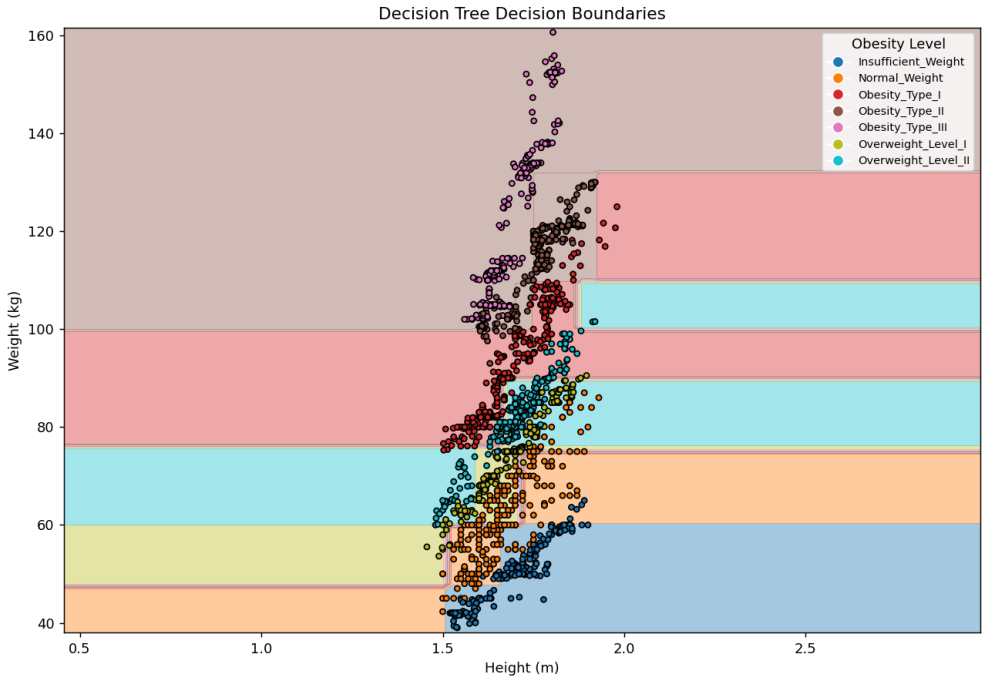


Figure 15. Decision tree decision boundaries on Height and Weight (leakage run).

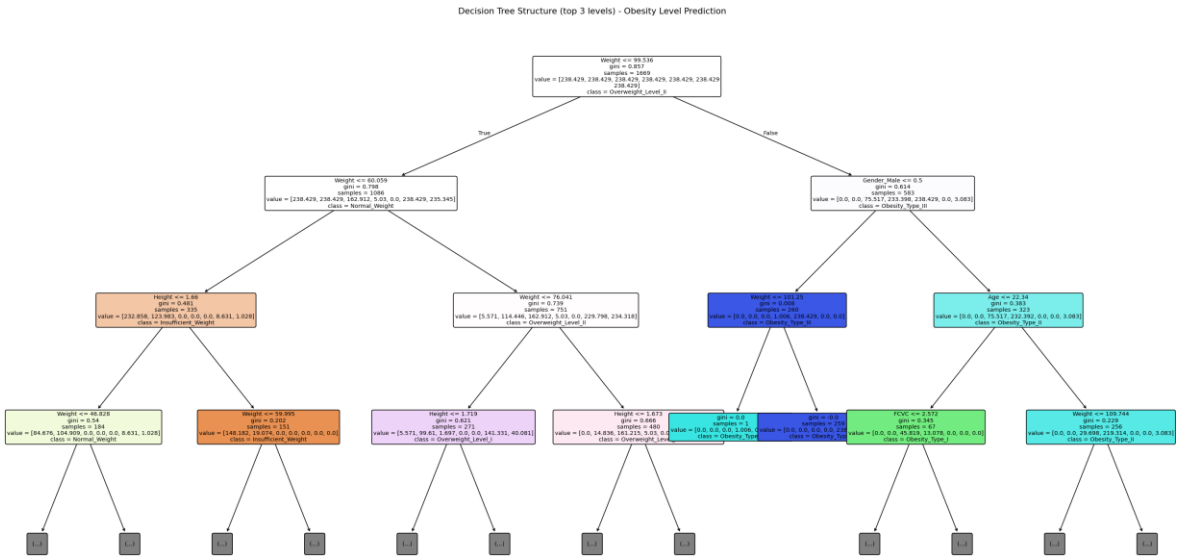


Figure 16. Decision tree structure, top three levels (leakage run).

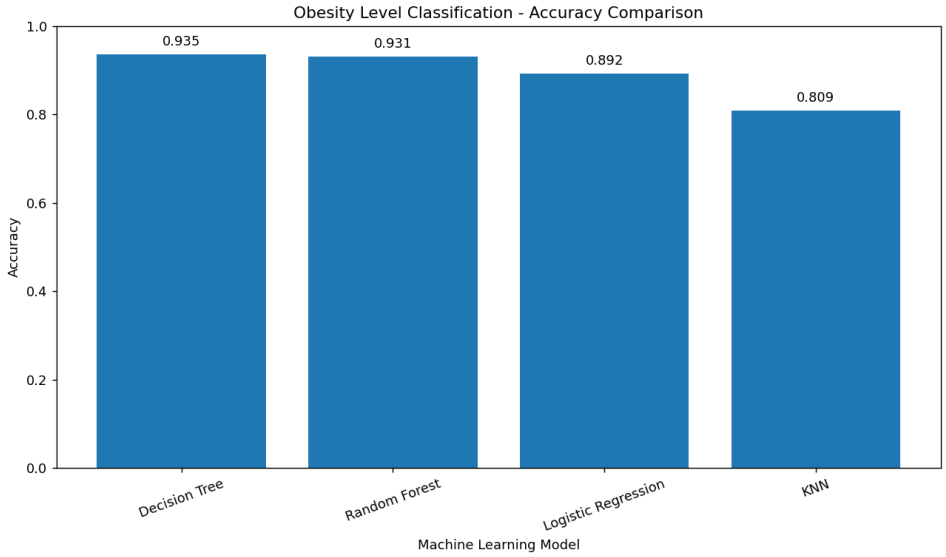


Figure 17. Accuracy comparison of the four models (leakage run).

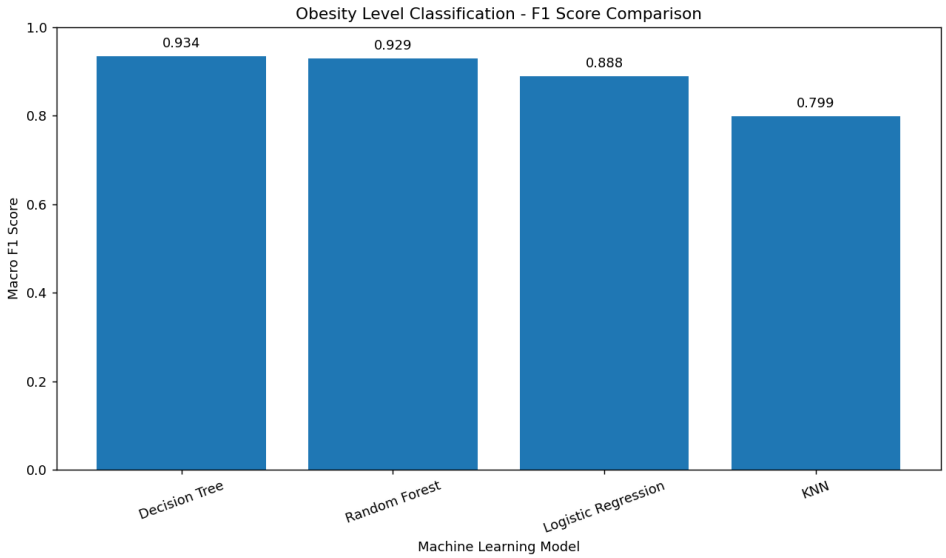


Figure 18. Macro F1 comparison of the four models (leakage run).

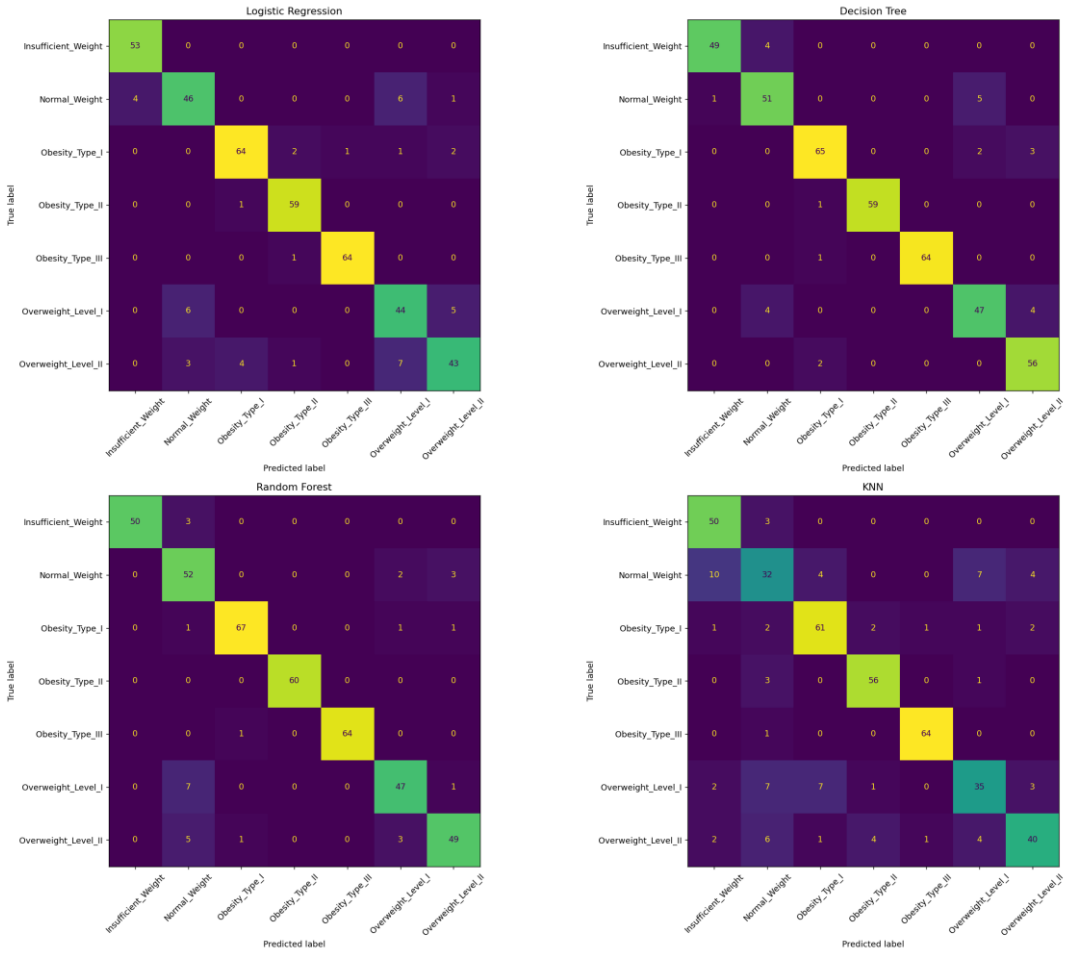


Figure 19. Confusion matrices of the four models (leakage run).

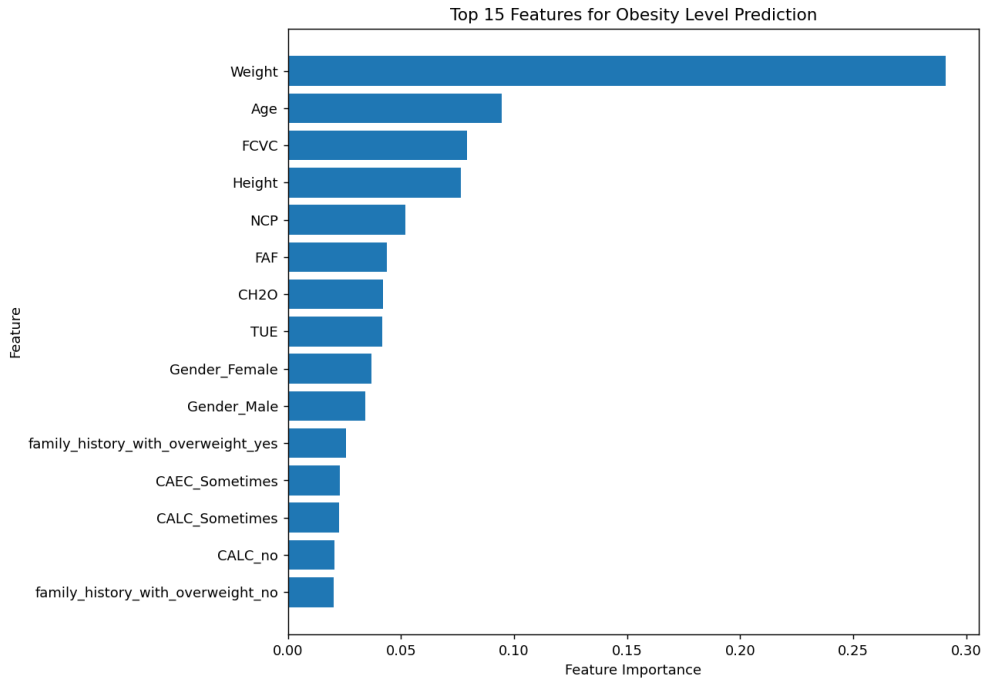


Figure 20. Top 15 Random Forest feature importances (leakage run).